

Compiling the Fly Brain

From Connectome to Computation: Extracting Algorithms from a Complete Nervous System

The fly-brain project

Working draft

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Introduction: What Does a Wiring Diagram Compute?

The question

A complete connectome — every neuron, every synapse — is now a real object you can hold in memory. For the male *Drosophila* central nervous system the object has 165,122 neurons and 104 million synapses. The natural question, and the one this book is about, is what it *computes*.

Not “what does it correlate with” — neuroscience has good tools for that — but the stronger claim: given only the wiring diagram, can we recover the algorithm? Can we say which populations form a continuous attractor, which compute a vector rotation, which implement a content-addressable memory — and then *prove it* in the only way a model can be proven, by predicting what happens when you break it?

The project this book documents attempts exactly that. The working metaphor is a **compiler**:

biology

- > canonical neural IR
- > motifs and structural signatures
- > dynamical primitives
- > competing executable hypotheses
- > validated computational models
- > compact algorithms / BioISA instructions
- > engineering implementations and benchmarks

The compiler analogy is chosen carefully. A real compiler does not understand a program by reading it once; it lowers source to an intermediate representation, runs analysis passes over the IR, and emits artifacts that can be executed and tested. Likewise here: the connectome is lowered to a species-agnostic IR, analysis passes detect candidate *operators* (computational primitives), and each operator is instantiated as an *ensemble* of executable models whose disagreements are resolved by predicted experiments.

The calibration target

The first circuit is the fly’s heading system in the central complex — the ring attractor formed by EPG, Delta7, PEG and PEN neurons. It is chosen for a specific reason: it is the best-understood circuit in any complete connectome. Physiologists already know it maintains a bump of activity

encoding heading; they have recorded it, perturbed it, and measured the consequences. That makes it a calibration instrument:

Can the machinery recover a known computation from structural constraints and limited functional evidence, without the answer being written into the extraction rules?

If it cannot recover this circuit, it cannot be trusted anywhere. If it can — and can also say *where it is uncertain* — the same machinery applies to the operators nobody understands yet.

The desired output

Not a simulation. A statement like this one:

Across plausible parameterizations, this circuit consistently supports a low-dimensional heading state. Two mechanisms remain compatible with the observations. Perturbing this cell population under this stimulus distinguishes them.

Three kinds of agreement are tracked separately throughout, because conflating them is the classic error in computational neuroscience:

- **Task-performance agreement** — models that solve the same problem.
- **Neural-response agreement** — models whose population activity matches.
- **Perturbation-prediction agreement** — models that break the same way.

Models routinely agree on the first while disagreeing entirely on the third. The whole methodology in this book is built to exploit that asymmetry: find the perturbation where surviving hypotheses diverge, rank it, and check it against real data.

What this book contains

The chapters follow the pipeline order:

- **Chapter 2** — the data and the canonical IR: what a connectome looks like as a data structure, and how the same loader handles the fly and the worm.
- **Chapter 3** — structure before dynamics: what the graph alone tells you, and the cross-species comparison that calibrates which numbers are unusual.
- **Chapter 4** — the operator layer: twelve detected computational primitives, their evidence, and their counterevidence.
- **Chapter 5** — the ensemble method: why wiring constrains a family of models, not a model.
- **Chapters 6–10** — the three circuits worked in detail: the ring attractor (in both a spiking ensemble and a continuous field model), the PFN→hΔB vector shifter, and the mushroom-body memory circuit.
- **Chapter 11** — the engineering benchmark: the decompiled estimator against standard algorithms, honestly matched.
- **Chapter 12** — the compiler machinery itself: declarative specs and the generic ensemble runner.
- **Chapter 13** — synthesis: what the three circuits jointly teach, and what the connectome provably cannot determine.

Honesty conventions

Three rules govern every claim in this book, and they are worth stating up front because they are unusual:

1. **Negative results are recorded, not tuned away.** When a mechanism fails under noise, that failure is a finding.
2. **Baseline comparisons are matched.** A biological estimator is never declared superior to a Kalman filter without giving the Kalman filter its fair parameters.
3. **Formalism disagreement is data.** When the spiking model and the field model disagree, the disagreement is published in the artifact and resolved against biology — not smoothed over.

The reader should hold this book to the same standard.

The Data and the Canonical IR

The object

The analysis runs on the male *Drosophila* CNS connectome (maleCNS): 165,122 neurons, 10,511,038 edges, 104,213,652 synapses, annotated with predicted neurotransmitters per neuron (acetylcholine, GABA, glutamate, dopamine, serotonin, octopamine, histamine, or *unknown* — and unknown is carried honestly rather than guessed).

Two properties of the data matter for everything downstream:

- **Edges carry synapse counts, not conductances.** A weight of 25 means “25 synaptic contacts were reconstructed,” not “this synapse moves the postsynaptic potential by X millivolts.” The mapping count→conductance is an assumption — probably the single most consequential one in the whole pipeline (Chapter 10 shows it failing informatively).
- **Neuron types are granular and structured.** KCg-m, KCab-s, MBON01–MBON35, PPL103, PEN_a/PEN_b, EPG. Populations are selected by type pattern, and hemispheric side (_L/_R suffixes) is preserved — the central complex analysis in Chapters 6–8 depends on it.

The canonical IR

connectome_ir.py lowers each connectome into a species-agnostic intermediate representation:

```
load_fly()    # maleCNS binary dump -> IR
load_worm()   # Cook 2019 C. elegans hermaphrodite CSVs + NT map -> IR
```

The IR is deliberately minimal — enough for graph-theoretic and dynamical analysis, nothing more:

field	content
N, edges, synapses	sizes
types	per-neuron type labels
meta	side, superclass, class annotations
nt	per-neuron predicted transmitter
CSR adjacency	indptr / indices / weights

Why an IR at all? Because the analysis passes — flow depth, motifs, sign structure, spectral properties — should not know or care which species produced the graph. The IR is what lets a *comparison* be meaningful rather than rhetorical.

Cross-species calibration

Running the same IR through the same passes gives the two animals side by side:

quantity	fly (165,122 cells)	worm (454 cells)
forward weight fraction	0.375	0.118
feedback weight fraction	0.124	0.717
lateral weight fraction	0.501	0.165
excitatory / inhibitory / unsigned	0.612 / 0.382 / 0.005	0.216 / 0.168 / 0.616
crossing (bilateral) weight	0.218	0.374
inhibitory fraction of crossing	0.442	0.158
cell types	11,752	169

Read carefully, this table is a small textbook in itself:

- **The worm is a feedback machine; the fly is a layered one.** 72% of worm synaptic weight sits on edges that point backward in the DAG ordering, versus 12% in the fly. The fly’s wiring is dominated by *lateral* edges (50%) — connections within a processing layer. Different animals, genuinely different graph regimes.
- **The worm’s sign structure is mostly unknown** (62% unsigned), the fly’s mostly known (0.5% unsigned). Any cross-species conclusion that depends on sign is weighted accordingly.
- **Crossing inhibition is qualitatively different.** In the fly, crossing edges are *more* inhibitory than ipsilateral ones (44% vs 38%) — consistent with bilateral comparison being done by inhibition. In the worm the pattern inverts.

This comparison does double duty. Scientifically it is interesting; methodologically it is the sanity check that proves the passes are species-agnostic rather than fly-shaped code that happens to run on the worm.

Where the IR stops

The IR answers “what is wired to what, how strongly, with what sign.” It cannot answer “what does it do” — that requires dynamics, which is the subject of Chapter 5 onward. The boundary is deliberate and it is the book’s central methodological commitment: **structure first, then dynamics, and never let the dynamics stage silently re-import structural assumptions.**

One concrete example of that discipline: the transmitter table maps each synapse’s effect sign from the *presynaptic* cell’s predicted transmitter, with *unknown* yielding no effect rather than an invented sign. When APL’s entry was found to read **undefined** in an early diagnostic, the correct response — documented in Chapter 10 — was to trace the sign path rather than to patch it.

Structure Before Dynamics: What the Graph Tells You

Before any neuron fires in any model, the wiring diagram already contains a great deal of computation-shaped information. `algo_ir.py` runs the generic passes; this chapter surveys what they found and — equally important — where structure stops being sufficient.

Flow: the brain is a layered machine with deep recurrence

Ordering the DAG condensation (304 iterations to reach a stable topological layering) splits the fly’s synaptic weight into:

- **50.1% lateral** — within-layer connections. Half the brain’s wiring is horizontal.
- **37.5% forward** — the sensory→motor direction.
- **12.4% feedback** — backward edges, a small but strategic minority.

Sensory neurons sit at hop 0; descending neurons are reachable in 1–3 hops from sensory input — the fly is *shallow* in the feedforward direction. Whatever long processing it does is done by recurrence, not depth. This is the single most important global fact for interpreting everything else: **computation lives in the lateral and feedback channels.**

Sign: inhibition is nearly half the budget

61.2% of synaptic weight is excitatory, 38.2% inhibitory, 0.5% unsigned. Inhibition is not a correction term in this brain — it is a primary computational resource. The bilateral data sharpens this: edges that cross the midline are 44% inhibitory versus 38% for ipsilateral edges, and there are 25 pairs of left/right homologues in mutual inhibition — the wiring pattern for hemispheric competition and comparison.

Motifs: a canonical vocabulary

At the cell-type level (11,752 types, 473,523 strong type-level edges), the reciprocity census gives:

- **20,816 excitatory inhibitory reciprocal pairs** — the dominant reciprocal motif is *signed*: a type excites another and is inhibited back. That is the wiring of feedback gain control, repeated twenty thousand times.
- 8,272 E E and 6,375 I I pairs, and only 786 mixed.

- The top recurrent excitation pairs concentrate in the optic lobe (L2 L4, L2 L5, Mi1 T2 — the motion pathway’s front end) and the mushroom body (DPM KCg-m, KCg-m PPL103 — memory and dopaminergic feedback).

Convergence/divergence asymmetries are extreme: sensory and intrinsic types fan out to tens of types (median out-degree 49 for VNC sensory) while motor neurons fan in (median in-degree 73 for VNC motor, out-degree 0). The graph’s own vocabulary already sketches the functional direction.

Hubs and spectral structure

Degree distributions are heavy-tailed (in-degree tail exponent ~ 1.0 , out-degree ~ 1.15). The dominant eigenvector of the weighted graph has participation ratio ~ 50 — a small set of neurons carries the dominant mode — and 32 eigenvalues have positive real part: the signed wiring is *linearly unstable* at many modes, i.e., activity in the real network is sculpted by nonlinearity and inhibition, not by passive decay.

Region structure

The per-region passes find the canonical specializations:

- **Optic lobe** — 47 columnar types, with retinotopic weight-sharing: the same input-weight profile repeated across columns (a convolved front end, Chapter 4).
- **Mushroom body** — 4,064 KCs fed by 282 projection neurons (expansion ratio ~ 6), each KC sampling a median of ~ 56 PN inputs through ~ 3 claws; APL covers 100% of KCs and is driven by 100% of them — the textbook random-projection-plus-normalization architecture.
- **Central complex** — the columnar ring analyzed in Chapters 6–8; the measured Delta7 kernel fits a cosine at $R^2=0.99$.

The ceiling of structure

Everything above is obtained without simulating a single spike. But a recurring result of the dynamical chapters is that **structure systematically overstates what dynamics delivers**: wired kernels are realized, wired phase shifts are realized at a third of their amplitude, wired attractors appear only in narrow gain regimes. The graph is a hypothesis generator of high quality and limited authority — the rest of the book is about how to test what it proposes.

The Operator Layer: A BioISA

`operators.py` scans the IR’s structural report and emits the project’s central artifact: a catalogue of candidate **computational operators** — the BioISA, in the project’s metaphor, an instruction set the nervous system appears to implement. Each entry carries a substrate (which cell types), a signature (what wiring pattern triggered the detection), semantics (input/state/output), evidence and counterevidence, and a confidence score.

The twelve detected operators:

operator	region	confidence	substrate / signature
ring_attractor	central complex	0.70	EPG, Delta7, PEG, PEN (88 cells); columnar ring + 2-hop inhibitory kernel $1-\cos$ + mirror-symmetric shifter
winner_take_all	motif	0.60	reciprocal-inhibition type pairs
phasor_vector_shift	central complex	0.55	PFN $\rightarrow$ h Δ B; consistent $-3/+2$ column offsets
motion_correlator	optic lobe	0.55	T4/T5 dendritic input arms; Reichardt-style spatial pairing
convolution_front_end	optic lobe	0.55	columnar weight-sharing across 40+ types
sparse_associative_memory	neuromuscular body	0.55	4,064 KCs, $\sim 6\times$ expansion, APL feedback, DAN $\rightarrow$ MBON teaching
feedforward_inhibition	motif	0.55	signed reciprocal pairs
divisive_normalization	antennal lobe	0.50	local interneuron feedback over PN pool
command_funnel	descending	0.30	extreme fan-in onto DNs

operator	region	confidence	substrate / signature
(plus low-confidence / generic re-detections of the same motifs)		0–0.25	

What a detection means — and does not

A detection is a *signature match*, not a function claim. `ring_attractor` scores 0.70 because its signature is unusually specific: a columnar population whose two-hop inhibitory kernel fits a cosine at $R^2=0.99$, plus a mirror-symmetric pair of shifted feedback populations. That combination essentially only means “continuous attractor with velocity input.” `command_funnel` scores 0.30 because “lots of fan-in” is consistent with many computations.

Every entry also records **counterevidence** and **predictions** — the detector is required to say what would prove it wrong. The mushroom-body entry, for example, predicts that KCs should be sparse and decorrelated and that APL should implement gain control; Chapter 10 shows what happened when that prediction was tested.

Signatures vs semantics

The catalogue maintains a deliberate split:

- **Signature** — what the wiring looks like (a cosine kernel, a column offset, an expansion ratio). Structural, measurable, reproducible.
- **Semantics** — what the circuit computes (a heading phase, a rotated vector, an odor identity). A hypothesis to be established dynamically.

`topology_folded` flags cases where the graph-theoretic form hides a geometric one — e.g., the ring attractor’s recurrent connections are *not* folded: the ring is a real ring in the wiring, not an artifact of the ordering.

From operator to experiment

Each operator entry is the input to the pipeline’s executable stage: the substrate names the populations to instantiate, the free quantities (gains the wiring doesn’t fix) become ensemble axes, and the counterevidence predictions become perturbations. Chapters 6–10 trace three operators end to end; Chapter 12 describes the machinery that makes a fourth operator a matter of writing a spec entry rather than a new script.

The Ensemble Method

Why ensembles

A connectome underdetermines a model in a specific, enumerable way. The wiring tells you the graph — which cells connect, how many synapses, what sign the transmitter implies. It does not tell you:

- the conductance each synapse produces (the count→PSP calibration);
- the strength of gap junctions (invisible in the EM data);
- neuromodulatory state;
- threshold, adaptation, and resting-potential heterogeneity.

So the honest object to construct is not *a* model but **the set of models consistent with the wiring** — an ensemble spanning the free parameters — and then to ask which conclusions survive across it. This inverts the usual logic of simulation: instead of tuning one model to fit data, we let structure define a family and measure where the family’s members agree and where they diverge.

The divergence is the product. If two ensemble members agree on baseline behavior but disagree about what happens when population X is silenced, then “silence X” is a *discriminating experiment* — and it can be ranked against all others by how much it splits the ensemble.

The machinery

`lif_ensemble.mjs` provides the substrate:

- `makeBuilder(D, edges, axes)` returns a factory: parameter values → a LIF network over exactly the circuit’s cells, with named edge classes scaled by the free parameters.
- `run(net, ms)` advances the event-driven simulation.
- `seedRng(seed)` makes every probe (odor patterns, noise, initial conditions) reproducible — ensemble comparisons are only meaningful at fixed randomness.

The LIF model itself (`src/lif.js`) is calibrated on published PSP sizes (Shiu et al. 2024): synaptic weights are count-scaled PSPs, sign comes from the presynaptic transmitter (GABA/glutamate inhibitory, acetylcholine and modulators excitatory, **unknown contributes nothing**), and a conductance-based option (`coba`) is available where reversal potentials matter.

The protocol

Each lab follows the same six-step loop:

1. **Grid** — the free axes are swept (e.g., {epgGain × penGain × d7Gain × epgTonic} = 48 members; {kc2mb × aplGain × mbonTonic × mbRecur} = 18 members).
2. **Probe** — each member is driven and measured with geometry-appropriate observables (bump phase/width on a ring; offset column vectors on a linear map; pattern separation and compression on a memory).
3. **Classify** — members are labeled by behavior class (silent, attractor_tonic, filter; wired_shift, passthrough; gain_controlled, linear_passthrough, collapsed).
4. **Perturb** — the same probe is repeated under each perturbation (silence a population, remove an edge class, scale a drive).
5. **Split** — for each experiment, the *separation score* counts how many member pairs it places in different outcome classes.
6. **Rank** — experiments are ordered by separation; the top experiment is the one where the surviving hypotheses diverge most.

The output is a JSON artifact: hypothesis counts, mechanism counts, the ranked experiment table, and a prose claim — e.g.:

Wiring supports a low-dimensional heading bump in a narrow gain regime (EPG recurrence ~3–6×, tonic EPG bias ~3–7 mV). Free-running bump without tonic drive was not observed at any grid point. Among bump-sustaining models, Delta7 silencing separates mechanisms that are indistinguishable at baseline.

Note the grammar of that claim: it is about the *ensemble*, not a model — it reports a regime, a non-observation, and the experiment that discriminates. That phrasing is the pipeline’s deliverable.

What the scores mean

Separation is an information-theoretic quantity wearing simple clothes: for an experiment that sorts 48 members into outcome classes, the score is high when the class assignment is informative about which member produced it. `tonic_sweep` scored 0.414 — the best discriminator in the ring lab — because members in different mechanism classes answer the sweep with categorically different bump behavior.

The same machinery runs across seeds; conclusions are only trusted when the *ranking* — not just the scores — survives reseed.

The Heading Circuit: A Spiking Ensemble

The first operator taken end-to-end is `ring_attractor` — the best-understood computation in any connectome and therefore the calibration target for the whole method.

The circuit

88 cells in five populations:

- **EPG** (46) — the ring population; bump position encodes heading.
- **Delta7** (42) — all-to-all inhibition among EPGs with a cosine profile.
- **PEG** (18) — EPG recurrence onto itself, offset by one column.
- **PEN_a** / **PEN_b** (20 + 22) — mirror-symmetric shifter populations, each wired to EPG with a ± 3 -column offset; carry the angular-velocity signal.

The measured two-hop EPG→Delta7→EPG kernel fits $a - b \cdot \cos(\Delta)$ with $R^2=0.99$ and contrast 0.878 — inhibition minimal at the bump, maximal opposite. Structurally this is the Amari ring recipe almost verbatim; the question is whether the *dynamics* deliver what the structure advertises.

The ensemble

`hypothesis_lab.mjs` sweeps four free axes — EPG recurrent gain, PEN gain, Delta7 gain, and a tonic EPG bias — over 48 members, drives each with a velocity profile plus a landmark cue, and measures the bump: existence, width, velocity tracking, and survival under perturbation.

The finding: a narrow attractor regime

class	members	meaning
<code>silent</code>	30	no sustained bump at any drive
<code>attractor_tonic</code>	12	bump exists — only with tonic EPG bias ~3–7 mV
<code>filter</code>	6	activity follows input without self-sustained bump

Two conclusions the wiring alone could not give:

1. **The attractor is gated.** Free-running bump without tonic drive was not observed at *any* grid point — the wiring needs a depolarizing bias to cross into the bistable regime. The fly’s heading system is a structure that must be switched on, not one that is always on.
2. **Delta7’s role splits the surviving members.** Among the 12 attractors, silencing Delta7 produces three distinct outcomes — bump collapses (`d7_essential`, 3), bump survives but *widens* (`d7_confines`, 1), bump survives *sharp* (`d7_sculpts_sharp`, 8). Same baseline behavior, different perturbation answers.

The ranked experiments

Separation scores at seed 42:

experiment	score	discriminates
<code>tonic_sweep</code>	0.414	attractor vs filter vs silent — the regime boundary
<code>peg_lesion</code>	0.370	whether the offset recurrence sustains the bump
<code>d7_silence</code>	0.351	the three Delta7-mechanism classes
<code>unilateral_pen_R</code>	0.284	PEN arm gating (velocity direction)
<code>unilateral_pen_L</code>	0.120	asymmetric — the two PEN arms are not equivalent

The headline: the ensemble cannot distinguish the Delta7 mechanisms at baseline — everyone’s bump looks the same — but `d7_silence` and `tonic_sweep` split them completely. That is the discriminating-experiment logic working as designed: the wiring gives you a family; perturbations tell you which member the fly is.

What the ensemble could not say

The LIF members classified Delta7’s role as either essential or sculpting — none predicted the *survives-but-widens* outcome that (Chapter 8 will show) the real data supports. The ensemble is only as good as its mechanisms; the missing one — a second, non-Delta7 inhibitory channel — was found by the field model and by published biology, not by more grid points. The ensemble’s honest answer was “we don’t know,” expressed as a mechanism split rather than a false consensus.

The Continuous Field: Deriving the Mechanism

The LIF ensemble answers “does the wiring support an attractor.” A different question — “what mechanism moves the bump” — wants a different formalism: an **Amari neural field**, a PDE on the ring in which the bump is an emergent solution of a continuous equation, not a property of 46 discrete cells. `ring_pde.py` implements it.

The model

A ring of n units ($n=256$ default, 32 columns) carries an activity field $u(\cdot)$:

$$\partial_t u = -u + W_{d7} * f(u) + \text{recur} \cdot K_{\text{exc}} * f(u) + \text{pen_input} + \text{landmark_input} - \text{rn} \cdot \text{mean}(f(u))$$

where each term is measured, not invented:

- W_{d7} — the Delta7 kernel, the fitted $a - b \cdot \cos$ ($a=873$, $b=890$);
- K_{exc} — a narrow local-excitation kernel, the EPG/PEG recurrence term (the one free gain);
- pen_input — the velocity pathway, in several formulations (below);
- landmark_input — ring-neuron-style disinhibition anchoring the bump to a cue;
- $\text{rn} \cdot \text{mean}(f(u))$ — the second inhibitory channel, added after withheld-data correction (Chapter 8).

f is a saturating nonlinearity — without it, recurrent excitation runs away; biology has rate saturation, and the model must too.

Deriving the velocity gain

The PEN pathway shifts the EPG bump during rotation. Naively, the wiring says “shift by $\pm s$ columns” — but at what *gain*? Earlier versions tuned it by hand; the field model derives it.

A bump $u^*(\cdot - (t))$ subject to shifted feedback $v \cdot K_s * f(u)$ moves by translation-mode projection — projecting the perturbation onto the bump’s marginal (Goldstone) direction $u^{*'} :$

$$\begin{aligned} - \dot{u}^{*'} &= v \cdot u^{*'}, K_s * f(u^*) \\ \rightarrow \dot{u}^{*'} &= v \cdot u^{*'}, K_s * f(u^*) / u^{*'}, u^{*'} \end{aligned}$$

The inner products are evaluated on an *equilibrated* reference bump (a subtlety that mattered: calibrating on the unrelaxed seed profile gave a wrong gain and a frozen benchmark). The computed

`pen_gain` 0.27 then makes the drift rate a **derived property of the kernels**, deterministic and parameter-free — the theory predicts ± 3.7 per unit drive, and the model tracks it.

The mechanism comparison

Four PEN mechanisms were implemented and run through the noisy benchmark:

mechanism	form	noisy RMS	verdict
shifted	literal shifted-synapse feedback, single field	1.29	distorts the bump under fluctuating
shifted2	amplitude-coded second field	1.61	worse — a second attractor doesn't repair it
shifted3	position-coded: displaces a PEN bump, EPG is pulled toward it	0.60	works — and predicts an EPG–PEN phase offset
advect	spectral transport of the field (first-order equivalent of shifted feedback)	0.41	the bound the mechanisms approach

The important result is not the ranking — it is the **mechanism discrimination**. Amplitude coding fails under noise in every formulation; position coding works at half the idealized cost and, crucially, predicts a *measurable* signature the amplitude model does not: the EPG and PEN bumps should be phase-offset during rotation — which is exactly what the fly literature reports.

The residual gap between **shifted3** (0.60) and **advect** (0.41) is itself a finding: indirect transport costs ~50% over the theoretical bound — the fly pays for a real mechanism with lag. And **shifted**'s failure is a falsifiable modeling claim: the single-field reduction drops something the real two-population PEN circuit needs.

Why two formalisms

The LIF ensemble and the field model are not redundant — they answer different questions and disagree informatively (Chapter 8). The field model can express a graded bump-width curve; the LIF ensemble can only vote among classes. The LIF model respects the actual cell-level wiring (including PEN arm asymmetry the field model smooths over); the field model exposes the mechanism's continuous limit. Using both is the method.

Cross-Formalism Validation: When Models Disagree

The strongest test the project has produced so far came not from a new model but from a *disagreement between existing ones* — resolved against data the pipeline had not seen.

The perturbation suite

`perturb_pde.py` runs the LIF lab’s top-ranked perturbations on the field model:

- **d7_sweep** — graded Delta7 suppression, measuring bump width and amplitude R.
- **pen_both** / **pen_left** / **pen_right** — PEN arm gating.
- **exc_sweep** — the free excitation gain, mapping the bistability threshold.

Two results validated the machinery: the PEN arm test gave left-arm drift 0.96 rad in 2 s and right-arm drift exactly 0 — correct arm selectivity — and the excitation sweep located the bistable threshold at `epg_recur` 1.5 (the free parameter’s operating point is now measured, not assumed).

The Delta7 disagreement

The graded sweep produced a clean curve:

d7 gain	FWHM	R
1.0	90.5°	0.597
0.6	98.1°	0.605
0.3	110.1°	0.599
0.1	126.8°	0.571
0.0	~143°	~0.53

The bump **survives complete Delta7 block** — widened and weakened, but formed. This directly contradicted the LIF ensemble’s majority class (`d7_sculpts_sharp`, 8 of 12 attractors predicted a sharp surviving bump) and matched only its minority `d7_confines` class (1 member).

The withheld-data check

Turner-Evans et al. (2020) had already done the real experiment. Their result:

The E-PG population organizes into a single bump even if the output of the $\Delta 7$ neurons is reduced... the bump no longer reliably tracks the fly’s movements.

The real bump survives. Neither the LIF majority nor the original single-channel field model predicted this — the original field model *dissolved* the bump below $d7$ 0.3 because it attributed all surround inhibition to Delta7.

The correction

The fix was not parameter tuning — it was a missing mechanism the paper itself names: **other inhibitory sources** (GABAergic ring neurons) must shape the bump. The model was corrected by splitting inhibition into two channels:

- the structured Delta7 cosine surround;
- a shallow, activity-dependent ring-neuron term `-rn·mean(f(u))` that scales with total bump mass.

With `rn_gain 0.5`, `depth 0.5` the corrected field model reproduces the real result through the whole sweep: FWHM $90^\circ \rightarrow 143^\circ$ as Delta7 is fully blocked, bump always survives. The benchmark cost is honest — tracking precision moved from 0.41 to ~ 0.48 RMS — because matching reality costs the model a little performance. That tradeoff is itself a finding: the fly’s two-channel inhibition buys robustness to losing a pathway, at the price of precision it does not need.

What the loop proved

The sequence is the project’s design goal made concrete:

1. LIF ensemble splits into mechanism classes under perturbation.
2. Field model produces a quantitative curve the LIF could not express.
3. The two formalisms **disagree** — and the artifact records `agrees_with_lif_majority: false`.
4. Published biology resolves the disagreement — the field model was right, but for an incomplete reason.
5. The model is corrected with the missing channel the data demanded.

The disagreement was never smoothed over. `hypothesis_lab`’s JSON artifact now carries a `cross_formalism` block that records the PDE outcome, the LIF class it maps to, and the flag that the two disagreed. When a reader opens the artifact they see the models arguing — which is exactly what a hypothesis-testing pipeline should show.

The PFN→hΔB Circuit: A Wired Transform That Isn't

The second operator through the lab is `phasor_vector_shift`: PFN neurons project to hΔB with consistent column offsets (−3 for PFNd, +2 for PFNv per the structural histogram), suggesting the circuit implements a fixed coordinate rotation — the “vector shift” that transforms heading into the compass frame the fan-shaped body uses.

The first-pass lab (72 members over `{pfnGain, hdRecur, fbGain, column}`) measured whether the realised population response reproduces the wired offsets. The deepened workup added what the first pass missed: three dynamical mechanisms the wiring shows but a static-offset analysis ignores.

The connectome’s hidden structure

- **PFN self-recurrence** — 7,970 synapses within PFNd alone. The shift may be *generated* by dynamics, not inherited from wiring.
- **hΔB→PFN feedback** — 61 edges. A recurrent loop the offset model treats as absent.
- **hΔB recurrence** — the return pathway, previously binary (on/off), swept continuously 0–2.

The finding: the shift is not a constant fold

72 members, hypotheses `{wired_shift: 13, passthrough: 55, silent: 4}` — but the mechanism split is the result:

finding	implication
0 of 13 <code>wired_shift</code> members are <code>wired_only</code>	The −3-column wiring alone <i>never</i> produces the shift — dynamics always participates
column sweep never rigid	the transform is <i>warped</i> by position, not a clean linear shift
PFNv rarely realises +2	arm asymmetry (20 vs 40 cells) that the structural histogram hides
mechanism counts spread across <code>needs_hdb_recur</code> , <code>needs_pfn_recur</code> , <code>needs_hd2pfn_feedback</code> and combinations	the shift is a <i>hybrid</i> — wiring biases, recurrence and feedback complete it

The interpretation changed meaningfully from the first pass. “PFN→hΔB implements a wired phase shift” is wrong; the truth is “the wiring provides a bias that recurrence and feedback complete into a shift.” A different mechanism class entirely — and one that makes a different experimental prediction.

Ranked discriminators

experiment	score
amp_scaling	0.424
hd_recur_off	0.406
hd2pfn_off	0.316
measure_pfnd_offset	0.314
pfnv_silenced	0.312
sweep_rigidity	0.282

The newly added feedback loop is already a top-3 discriminator — removing hΔB→PFN feedback collapses the shift in members that need it. **amp_scaling** top-ranks because the phasor arms’ amplitude asymmetry (the 20-vs-40 cell imbalance) is itself mechanism-dependent.

Why this matters for the method

The phasor circuit is the project’s clearest demonstration that **structure overstates**: the wiring says “−3 shift,” the dynamics say “−3 is an upper bound that recurrence partially realizes and position-dependently warps.” A naive reader of the connectome would have published the wired offset as the circuit’s function. The ensemble says the function is a dynamical completion of a structural bias — and says *which perturbation* proves it.

The Mushroom Body: An Honest Negative

The third circuit is the stress test the compiler was built for — an operator with no spatial geometry at all. `sparse_associative_memory` targets the mushroom body: 4,064 Kenyon cells, the giant feedback interneuron APL, 97 MBON output neurons, and ~340 dopaminergic PPL/PAM cells. The computation is pattern separation and associative readout — nothing like a ring or a column map.

The circuit

The wiring (Chapter 3) reads like a textbook associative memory:

- **Expansion:** 282 PNs $\rightarrow$ 4,064 KCs ($\sim 6\times$ ratio), each KC sampling a median of ~ 56 PN inputs through ~ 3 claws — the random-projection step.
- **Normalization:** APL covers 100% of KCs and is driven by 100% of them — the classic divisive-normalization loop.
- **Readout:** KC $\rightarrow$ MBON (44k edges, median 313 KCs per MBON).
- **Teaching:** DAN $\rightarrow$ MBON in the same compartment (70% of DAN-MBON pairs).

The protocol

The `memory` geometry in `run_ensemble.mjs` drives two overlapping KC “odor” patterns (200 cells, 50% shared) directly, and measures:

- **separation / expansion** — cosine similarity between the two MBON output vectors relative to the KC input overlap;
- **compression** — KC spike count at $2\times$ drive vs $1\times$ drive (the gain-control probe);
- the ensemble sweeps `{kc2mb, aplGain, mbonTonic, mbRecur}` (18 members).

The result: gain control is not established by the wiring

18 members classified: 9 `silent`, 9 `collapsed/linear_passthrough` — **zero `gain_controlled`**. The direct diagnostic is decisive:

drive	KC spikes (400 KCs)	APL spikes
60	4,163	13, 8

drive	KC spikes (400 KCs)	APL spikes
150	9,144	16, 11
300	14,338	16, 11

APL **saturates** at ~16 spikes while KC output scales 3.4×. The feedback cannot compress because the recruitment saturates — and the structural reason is that APL→KC is only ~4,210 edges for 4,064 KCs: **roughly one synapse per KC**. At connectome-count weights, the divisive-normalization loop is too sparse to matter.

What this means — and what it does not

This is the lab working as intended, and the negative result is the finding:

- **The wiring does not establish gain control.** Whether the real MB normalizes depends on per-synapse conductance — a parameter the connectome simply does not carry. The ensemble correctly reports “insufficient evidence,” expressed as a missing hypothesis class rather than a tuned affirmative.
- **MBON readout is linear, as expected.** expansion < 1 for 50%-overlap odors is correct — decorrelation happens upstream (odor→KC), not at the readout. Driving KCs directly bypasses the layer where the memory’s actual computation lives, so the MBON transform being near-linear is the *right* answer, not a failure.
- **APL’s sign was a real bug risk.** Early diagnostics found nt: **undefined** for APL — which in this simulator means *edges contribute nothing*. APL turned out to be correctly GABAergic (2 cells, both **gaba**), so the loop is inhibitory as it should be; the weakness is quantitative, not a sign error. But the check was necessary — a sign bug would have produced the same null result invisibly.

The discriminating experiments

apl_silence and mb_recur_off are the perturbations; at the current parameterization they are nearly undiscriminating (~0.53 separation, essentially a coin flip across members) because APL is too weak for its silencing to matter. That is itself informative: **the experiment is only discriminating where the mechanism is present** — if a future calibration gives APL enough conductance to fire, the same perturbation becomes the top discriminator, and the ensemble will have predicted which measurement to make.

The lesson

The mushroom-body chapter is the book’s clearest statement of the project’s core discipline: the connectome constrains a *family*, and the family’s honest output includes “we cannot determine this from structure.” The sparse-memory operator is detected with confidence 0.55, its wiring is canonical, and its key mechanism is provably underdetermined — three facts that coexist without contradiction.

The Engineering Benchmark

A biological circuit that computes can be compared, fairly, against the algorithms an engineer would reach for. `bench_heading.py` does that for heading estimation: noisy angular-velocity input plus intermittent landmark observations, with occasional corrupt landmarks — the problem every navigation system solves.

The estimators

estimator	what it is
dead reckoning	integrate ; landmarks ignored
complementary filter	blend integrated and landmark at fixed
Kalman (1-D)	optimal linear estimator, tuned
Kalman + gate	same, with an outlier-rejection gate on landmarks
ring scalar	the scalar phase-and-population reduction of the attractor
ring field (PDE)	the full Amari field from Chapter 7
ring field + cell noise	the PDE with per-unit noise, $\sigma=0.15$

The matched-baseline rule is enforced: the complementary filter is run at the same effective correction strength as the biological estimator, so a win means something structural, not a tuned gain.

Results

estimator	RMS	post-outlier
dead reckoning	0.679	0.633
complementary $\sigma=0.35$	0.701	1.143
Kalman 1-D	1.117	1.991
Kalman + gate	0.489	0.710
complementary $\sigma=0.05$	0.604	0.633
ring scalar	0.641	0.668
ring field (PDE)	0.449	0.450
ring field + cell noise	0.453	0.455

The honest verdict

Three claims, in increasing order of interest:

1. **At the scalar level the biological estimator is not special.** Matched-strength complementary filtering ties the ring scalar (~ 0.60 – 0.64 both). The scalar algorithm the attractor implements is on the same tradeoff curve as the textbook ones.
2. **The advantage is in the spatial representation.** The PDE field beats every classical estimator including the gated Kalman — post-outlier 0.450 vs 0.710 . The mechanism is structural: a corrupt landmark must *win a spatial competition* on the field — nucleate a competing bump against antipodal inhibition — rather than shift a point estimate. Outlier robustness is free; there is no gating logic to tune.
3. **Per-cell noise costs almost nothing.** $\sigma = 0.15$ field noise degrades the PDE from 0.449 to 0.453 — the continuous representation is robust to the kind of noise that cripples scalar estimators.

What this does and does not claim

This is **not** “the fly beats the Kalman filter.” It is: *in this regime, with corrupt landmarks, a spatial-competition estimator is a competitive mechanism class* — and its robustness is architectural, not tuned. That is the correct way to state a biological engineering result, and it is the phrasing this book uses throughout.

The Compiler: Specs and the Generic Runner

The pipeline’s first version had a weakness the circuits themselves exposed: the three labs were hand-written scripts, each encoding its circuit’s populations, observables, and perturbations in bespoke code. That is “we ran three analyses,” not “the compiler compiles.” The fix is the missing middle layer.

Declarative specs

`ensemble_spec.py` emits a spec for each detected operator — the circuit-specific content as data:

```
{
  "geometry": "memory",          # which plugin interprets the observables
  "operator": "sparse_associative_memory",
  "populations": {"kc": "KC*", "apl": "APL*", "mbon": "MBON*", "dan": ["PPL*", "PAM*"]},
  "roles": {"input": "kc", "output": "mbon", "control": "apl"},
  "odor_size": 200, "odor_overlap": 0.5,
  "free": {"kc2mb": [...], "aplGain": [...], "mbonTonic": [...], "mbRecur": [...]},
  "perturbations": {"apl_silence": {...}, "mb_recur_off": {...}},
}
```

Populations resolve by type prefix (`KC*` → all Kenyon-cell types); roles name the functional slots; **free** lists the gain axes the wiring does not fix; perturbations are named interventions. The spec carries *provenance* — each axis is annotated with why it is free (count→conductance calibration, tonic bias unknown, etc.).

The generic runner

`run_ensemble.mjs` executes a spec against the connectome. It handles the parts that are circuit-independent:

- population resolution (exact and prefix patterns);
- the ensemble grid over free axes;
- the probe/perturb/rank loop;
- hypothesis classification and mechanism counting;
- JSON artifact output.

What is *not* generic is the observable geometry — how “did the computation happen” is measured. Three geometry plugins exist:

geometry	used by	observable
ring	ring_attractor	bump phase, width, angular velocity on S^1
linear	phasor_vector_shift	offset column vectors, phasor transforms
memory	sparse_associative_memory	pattern separation, gain compression

A fourth operator (e.g. `motion_correlator`, which needs synaptic delays and direction-selective pairing) will require a new plugin — and the design question becomes “what geometry class is this,” which is itself a meaningful classification.

What “compiled” now means

Adding the mushroom-body operator required: one spec entry (~30 lines of declarative data) plus one geometry plugin (~60 lines). The runner did everything else. That is the compiler claim made real — narrowly, but really:

The detected operator + measured structure → a declarative spec → an executed ensemble → a ranked experiment table, without circuit-specific code in the loop.

The remaining hand-authoring is the spec itself — the free axes are still chosen by a human who knows which parameters the wiring leaves open. Automating that (every unmeasured edge class becomes an axis; every substrate population becomes a perturbation) is mechanical and is the natural next step.

The artifact schema

Every run emits the same JSON shape: **seed**, **summary** (ensemble size, class counts, best experiment), **members** (per-parameter outcomes), **ranked_experiments**, and — where applicable — **cross_formalism** blocks. Downstream tooling (the web viewer, the report generator) consumes a uniform schema; a new circuit’s artifact is automatically consumable the moment it exists.

Synthesis: What Three Circuits Taught Us

The meta-finding

Across three circuits, three geometries, and two model formalisms, one result recurs:

Structure systematically overstates what dynamics delivers.

- The Delta7 cosine kernel fits the wiring at $R^2=0.99$ — and the attractor exists only in a narrow tonic regime the wiring does not reveal.
- The PFN $\rightarrow$ h Δ B wiring says “−3 column shift” — the dynamics realise it at $\sim 1/3$ amplitude, warped by position, and only with recurrent participation.
- The mushroom-body wiring shows the canonical APL normalization loop — and the counted synapses cannot carry it.

This is not a counsel of despair — it is a *calibration signal*. The gap between what structure advertises and what dynamics delivers is exactly the quantity that tells you where to look next: the missing tonic bias, the missing recurrence, the missing per-synapse conductance. Each gap is a specific, locatable demand for the data that would close it.

What the machinery can do now

The pipeline’s demonstrated capabilities, in order of increasing strength:

1. **Ensemble fitting** — wire-constrained model families, classified by behavior.
2. **Mechanism discrimination** — perturbations ranked by how completely they split the surviving hypotheses.
3. **Cross-formalism checking** — spiking and field models compared on the same perturbation, disagreement recorded.
4. **Withheld-data correction** — published biology used to fix a model, not just to validate it.
5. **Honest negative results** — “the wiring underdetermines this mechanism” as a first-class output.

What the connectome cannot determine

The failures are as catalogued as the successes:

- **Per-synapse conductance** — synapse counts are not PSPs. The mushroom-body chapter is the case study.
- **Neuromodulatory state** — the dopaminergic/octopaminergic layer is largely invisible to a count-based model.
- **Gap junctions** — absent from the EM reconstruction, present in the animal.
- **Temporal dynamics below the synapse-count scale** — delays, time constants, adaptation; the LIF carries one set of values where biology carries a distribution.

The right framing: these are not deficiencies of the method — they are the boundaries of the data, and the pipeline’s job is to *locate* them precisely. “APL gain control needs conductance values” is a more useful scientific output than either a false positive or silence.

Open problems, in priority order

1. **Auto-derive the spec axes.** Every unmeasured edge class $\rightarrow$ a free parameter; every substrate population $\rightarrow$ a perturbation. Removes the last hand-authored step.
2. **A fourth operator with a new geometry.** `motion_correlator` (T4/T5) needs synaptic delays; `divisive_normalization` (AL) re-tests the APL question on a circuit where the interneuron count is different.
3. **Quantitative withheld validation beyond the Delta7 curve.** The FWHM-vs- suppression curve is specific enough to check against imaging data point by point.
4. **Merging cross-formalism scores into the ranking itself** — experiment scores currently weight only within-LIF disagreement; a unified score would weight disagreements that survive both formalisms higher.
5. **The PFN $\rightarrow$ h Δ B loop closed** — the feedback perturbation is a top-3 discriminator; the experiment that would resolve it is now known and ranked.

The claim, restated

The connectome is a compiled artifact — the output of a developmental program that produced a machine whose description is a wiring diagram. This book documents a decompiler: a pipeline that reads the diagram, proposes the computations, builds the model families, and identifies the experiments that decide between them. It works where it should (the ring attractor is recovered and corrected against real data), fails where it should (the mushroom body’s missing parameter is located precisely), and reports the difference honestly throughout.

That is what “what have we got” currently means.

Appendix: Reproduction

Every number in this book is generated by a committed script and a committed artifact. The pipeline is deterministic (seeded RNG throughout); rerunning any stage reproduces the cited table.

Pipeline order

```
# IR and generic structure (Ch. 2-3)
python3 scripts/connectome_ir.py      # fly + worm -> canonical IR
python3 scripts/algo_ir.py            # IR -> generic structure report
python3 scripts/check_structures.py   # fly-through-IR reproduces algo_structures.json

# Operator detection (Ch. 4)
python3 scripts/operators.py          # structure report -> public/data/operators.json

# Ensembles (Ch. 5-6, 9-10)
python3 scripts/ensemble_spec.py      # -> public/data/ensemble_specs.json
node scripts/run_ensemble.mjs ring_attractor
node scripts/run_ensemble.mjs phasor_vector_shift
node scripts/run_ensemble.mjs sparse_associative_memory
# (the original hand-written labs: hypothesis_lab.mjs, phasor_lab.mjs)

# Field model and perturbations (Ch. 7-8)
python3 scripts/ring_pde.py           # self-check: bump, integration, landmark remap
python3 scripts/perturb_pde.py        # -> public/data/perturb_pde.json

# Benchmark (Ch. 11)
python3 scripts/bench_heading.py
```

Key artifacts

artifact	content
public/data/algo_structures.json	full structural report (fly)
public/data/ir_generic.json	cross-species IR comparison
public/data/operators.json	the BioISA catalogue
public/data/hypothesis_lab.json	ring ensemble, ranked experiments, cross-formalism block

artifact	content
public/data/phasor_lab.json	PFN $\rightarrow$ h Δ B ensemble
public/data/sparse_associative_memory_lab.json	memory ensemble
public/data/perturb_pde.json	field-model perturbation curves
public/data/ensemble_specs.json	the declarative specs

Notation conventions

- Ensemble “members” are parameter points; “classes” are behavior labels assigned by the probe; “mechanisms” are the perturbation-outcome subclasses.
- Separation scores are ensemble-splitting measures in $[0,1]$; a score near 0.5 at small ensemble size is near-maximal.
- All bump widths are FWHM in degrees; all benchmark scores are radians RMS.
- “Wired” refers to quantities read from synapse counts; “realised” to quantities measured in simulation.